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碩士論文

使用多尺度語意標記和輕量化 **Bi-LSTM** 模組實作基於歌詞的音樂結構分析

Multi-Scale Semantic Tagging for Lyrics-Based Music Structure Analysis Using Lightweight Bidirectional Long Short-Term Memory

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摘要

自動音樂結構分析在音樂資訊檢索領域中扮演著至關重要的角色，然而目前最先進的音訊模型（如 SongFormer）通常需要龐大的計算資源，限制了其在邊緣設備部署的適用性；為了解決此限制，本研究探索基於歌詞的方法是否能夠提供具競爭力的結構特徵線索。

本研究設計了一個基於優化雙向長短期記憶網路（Bi-LSTM）架構的輕量化且具音樂感知能力之歌詞分析框架，提出的模型結合了階層式 RoBERTa 嵌入向量、位置資訊與重複性特徵線索，以增強結構表示學習能力，在 DALI 資料集與 Harmonix Set 對齊子集上的實驗結果顯示，本研究方法展現出優異的有效性，將 LongFormer 的加權 F1 分數從 0.5672 提升至 0.7848；此外，本研究所提出之歌詞框架在核心樂段（包含主歌、導歌、副歌、橋段）的表現上，能與 SongFormer 達到具競爭力的效能，同時降低了 20.5 倍的 GPU 記憶體消耗。

實驗結果突顯本方法在結構分類精準度與計算效率之間的優異平衡，顯示基於歌詞的表示法為受限資源環境下的高效音樂結構分析提供了一個具潛力的發展方向。

關鍵詞：音樂結構分析、音樂資訊檢索、雙向長短期記憶網路、歌詞分析、深度學習

Abstract

Automatic music structure analysis plays a crucial role in music information retrieval. However, state-of-the-art audio-based models such as SongFormer require substantial computational resources, limiting their suitability for edge deployment. To address this limitation, we explore whether a lyrics-based approach can provide competitive structural cues.

We therefore design a lightweight and music-aware lyrics analysis framework built upon an optimized Bi-LSTM architecture. The proposed model integrates hierarchical RoBERTa embeddings with positional and repetition cues to enhance structural representation learning. Experiments conducted on synchronized subsets of the DALI dataset and Harmonix set demonstrate the effectiveness of the approach, improving the weighted F1-score of LongFormer from 0.5672 to 0.7848. Furthermore, the lyrics-based framework demonstrates competitive performance compared with SongFormer across core sections—including Verse, Pre-Chorus, Chorus, and Bridge—while reducing GPU memory consumption by 20.5×. Our results highlight a favorable balance between structural accuracy and computational efficiency, suggesting that lyrics-based representations provide a promising direction for efficient music structure analysis in resource-constrained environments.

Keywords: Music Structure Analysis, Multimedia IoT, Resource-efficient Machine Learning, Section Classification

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Chapter 1 Introduction

Music Structure Analysis (MSA) aims to segment a musical piece into semantically meaningful sections such as *Verse*, *Chorus*, and *Bridge*. As a fundamental task in Music Information Retrieval (MIR), MSA supports applications including music genre classification, personalized recommendation, music retrieval, and AI-assisted music production. Accurate structural segmentation enables better temporal understanding, semantic indexing, and efficient organization of large-scale music collections. Although recent advances have significantly improved MSA performance through deep learning, achieving both accurate structure prediction and computational efficiency remains a challenging problem.

Most existing MSA systems rely on acoustic representations derived from audio signals. While recent transformer-based approaches enable full-song modeling, these methods typically require high-dimensional spectrogram inputs and large model architectures, resulting in substantial computational and memory overhead. Processing a full-song spectrogram may involve extremely long sequences (on the order of 10^5 tokens) and require several gigabytes to tens of gigabytes of GPU memory, limiting their practicality in resource-constrained environments.

From a systems perspective, this limitation is particularly critical in emerging Multimedia Internet of Things (IoMT) scenarios, where devices are often constrained by bandwidth, memory, computational capability, and energy consumption. Transmitting or processing raw audio signals is inherently expensive due to their high data rate, making audio-based models less suitable for edge deployment. In contrast, lyrics provide a highly compact modality, typically consisting of only a few hundred tokens per song, offering orders-of-magnitude reductions in both data size and computational demand.

This lightweight representation makes lyrics an attractive alternative for scalable and real-time music understanding in IoMT systems.

Beyond computational efficiency, lyrics inherently encode rich semantic and structural information. Repeated lyrical phrases frequently correspond to recurring musical sections such as the Chorus, while narrative progression often reflects the transition between different functional sections. Consequently, lyrics provide strong structural cues that can complement or even substitute acoustic information for music structure prediction.

Recently, lyrics-based approaches have begun to adopt powerful pre-trained language models, such as LongFormer, to model long-range dependencies within song lyrics. However, these methods largely treat lyrics as conventional natural language sequences and rely on general-purpose language representations to implicitly learn musical structures. Unlike ordinary documents, song lyrics exhibit distinctive musical regularities, including semantic repetition, hierarchical organization, and strong positional dependencies that closely correspond to functional song sections. These music-specific characteristics are generally not explicitly modeled by existing NLP architectures, limiting their ability to fully exploit the structural information unique to lyrics.

This observation motivates us to rethink lyrics-based music structure analysis from a music-aware perspective. Instead of relying solely on increasingly larger language models to implicitly discover musical regularities, we investigate whether explicitly incorporating lyrics-specific structural cues can provide stronger inductive bias for music structure prediction while maintaining a lightweight architecture suitable for resource-constrained environments.

To address these challenges, this work proposes a lightweight lyrics-based framework for music structure prediction. We focus on English popular music produced after the 1950s, where lyrical organization closely correlates with musical form. Experiments are conducted on the intersection of the Harmonix Set [1] and the DALI dataset [2], producing a synchronized corpus with aligned lyrics, timestamps, and structural annotations. Based on this synchronized dataset, we propose a music-aware feature representation that integrates hierarchical semantic embeddings with explicitly modeled structural priors, including repetition-aware similarity and positional cues. These representations are subsequently modeled using an efficient Bidirectional Long Short-Term Memory (Bi-LSTM) network for structural section prediction.

The main contributions of this work are summarized as follows:

- **Synchronized Dataset Construction:** We construct a high-quality synchronized dataset by intersecting DALI v2.0 with the Harmonix Set, producing aligned annotations—including line-level lyrics, timestamps, and structural section labels—paired with corresponding audio files and spectrograms. This dataset provides a synchronized evaluation corpus, enabling consistent and fair evaluation between lyrics-based and audio-based music structure analysis methods.

- **Efficient Music-Aware Lyrics Framework:** We propose a lyrics-oriented feature representation that explicitly models hierarchical semantic context, repetition-aware similarity, and positional cues. Integrating these music-specific structural inductive biases into a lightweight Bi-LSTM architecture, we achieve a 20.5× reduction in peak GPU memory usage (189.3 MB vs. 3890 MB) and cut training time per epoch from 6.6 to 1.9 minutes, demonstrating its high efficiency for resource-constrained IoMT environments.

- **Competitive Cross-Modal Performance:** Despite its lightweight and efficient design, the proposed lyrics-based framework achieves performance comparable to or exceeding state-of-the-art lyrics-only and audio-only methods across major structural categories while requiring substantially lower computational resources.

Chapter 2 Related Works

2.1 General Music Structure Analysis

Music Structure Analysis (MSA) is a fundamental task in Music Information Retrieval that aims to decompose a music track into functional segments such as verse, chorus, and bridge. Early approaches primarily relied on signal processing techniques and hand-crafted acoustic features to identify structural boundaries. Methods based on self-similarity matrices (SSM) and novelty detection were widely used to capture repetition and transitions by analyzing timbral and harmonic patterns [3].

With the development of machine learning, supervised approaches have been introduced to improve structural modeling. Convolutional Neural Networks (CNNs) [4] have been applied to SSM representations for boundary detection, while Recurrent Neural Networks (RNNs) model temporal dependencies across musical segments. These methods learn discriminative representations from data and outperformed rule-based systems in segmentation accuracy.

Despite these advances, accurately modeling long-range structural dependencies and global song organization remains challenging, particularly when relying solely on acoustic cues. Recent multitask frameworks have attempted to mitigate this by jointly modeling metrical and functional structures to provide stronger inductive biases [5].

2.2 Audio-Based Music Structure Modeling

Recent progress in self-supervised learning (SSL) has led to powerful audio representation models. Frameworks such as MuQ [6], and MERT [7] learn general-purpose music embeddings from large-scale unlabeled audio corpora, enabling strong performance across downstream MIR tasks [8, 9]. Building on these representations, transformer-based architectures such as SongFormer [10] have been proposed to capture long-range dependencies for music structure analysis.

However, these audio-based approaches typically require high-resolution spectrogram inputs and large model capacities. Processing full-length songs often incurs substantial computational cost in terms of memory usage and floating-point operations, which can limit their applicability in real-time systems or resource-constrained environments. This motivates the exploration of alternative modalities that provide structural cues with lower computational overhead.

2.3 Lyrics-Based Music Structure Modeling

Lyrics provide a complementary modality for music structure analysis by encoding semantic information and textual repetition patterns. Repeated lyrical phrases often correspond to structurally important sections such as choruses, making text a useful signal for segment identification.

Earlier approaches relied on traditional natural language processing techniques, including TF-IDF and bag-of-words representations combined with probabilistic models such as Hidden Markov Models (HMM) and Conditional Random Fields (CRF). While effective at modeling local transitions, these methods struggle to capture long-range semantic dependencies and the figurative language commonly found in song lyrics.

Recent work has shifted toward dense representations and transformer-based architectures, such as LongFormer [11], to better handle long textual sequences. Nevertheless, many existing models rely on general-purpose language representations and do not explicitly incorporate music-specific structural characteristics, such as semantic repetition and positional regularity.

In addition, while sequence labeling frameworks such as Bi-LSTM-CRF can enforce label consistency, the associated decoding procedures introduce additional computational overhead. These limitations suggest a gap between expressive modeling and efficiency, particularly for lightweight deployment scenarios.

To address this gap, our approach focuses on explicitly modeling semantic repetition through a dedicated three-dimensional similarity feature (F_{sim}), integrated within an optimized Bi-LSTM architecture. By incorporating both semantic similarity and positional cues, the proposed framework captures structural regularities while maintaining a favorable balance between accuracy and computational efficiency.

2.4 Multimodal Audio–Lyrics Modeling

Recent studies have explored multimodal approaches that jointly model audio and lyrics to improve music understanding. For example, prior work has combined acoustic and textual features for emotion recognition [12,13], while models such as CLaMP [14] and MusicLIME [15] leverage cross-modal learning for retrieval and interpretability.

Although multimodal methods often achieve strong performance, they typically require synchronized access to both audio and lyrics, increasing system complexity and limiting applicability in scenarios where one modality is unavailable. Furthermore, most existing work focuses on tasks such as emotion recognition or retrieval, rather than structural segmentation. Accurately identifying structural functions, such as identifying a chorus or verse [16], is critical for music understanding.

Consequently, whether lyrics-only approaches can provide sufficient structural information for accurate music structure analysis remains an open and relatively underexplored problem. This work aims to investigate this direction by developing a lightweight, lyrics-based framework for structure modeling.

2.5 Lyrics-Informed MIR and the Proxy Paradigm

Traditionally, lyrics have been utilized as a complementary source to enhance audio-based models, a field known as lyrics-informed Music Information Retrieval (MIR). A common application is audio-lyric alignment, where forced alignment techniques [17] or attention mechanisms are used to synchronize textual tokens with acoustic frames to improve segmentation and retrieval accuracy. Recent advances in this field have further explored cross-modal semantic alignment to bridge the gap between symbolic lyrics and acoustic features [14].

However, most lyrics-informed systems treat text as a secondary modality that must be grounded in high-bandwidth audio signals. Our work shifts this paradigm by treating lyrics as a *lightweight proxy* for musical structure. By demonstrating that semantic and repetition patterns in lyrics alone can capture functional boundaries—often with higher precision than audio-driven models in transitional sections—we highlight a path for MSA that bypasses the need for resource-intensive acoustic processing. This lyrics-based approach is particularly vital for scenarios where raw audio transmission is restricted by privacy or bandwidth constraints.

2.6 Efficient NLP for Resource-Constrained Environments

Deploying large language models on edge devices has become an important research direction for resource-constrained IoMT applications. Although the pre-trained RoBERTa-base model provides rich semantic representations, its computational and memory requirements may limit deployment on low-power devices [18]. Recent advances in efficient NLP have introduced distilled language models such as DistilBERT [19] and TinyBERT [20], which substantially reduce model size and inference cost while preserving most of the performance of their teacher models.

Our framework is designed with modular scalability in mind. Since semantic feature extraction is decoupled from the lightweight Bi-LSTM classifier, the RoBERTa encoder can be readily replaced with distilled alternatives without modifying the remaining architecture. Although this study adopts RoBERTa-base to maximize semantic representation quality, the proposed framework naturally accommodates more efficient encoders in future work, enabling a flexible trade-off between accuracy and computational efficiency. Compared with transformer-based audio models such as SongFormer, which process substantially longer audio sequences with quadratic attention complexity, the proposed lyrics-based framework requires considerably lower memory and computational resources, making it a promising solution for edge-oriented music structure analysis.

Chapter 3 Dataset and Pre-processing

3.1 Data Integration and Dataset Synchronization

A critical challenge in Music Information Retrieval (MIR) research is the scarcity of ”triple-synchronized” data—recordings that simultaneously provide high-quality audio, precise line-level lyric timestamps, and expert-validated structural labels. While large-scale lyric corpora exist in the public domain, they typically lack the temporal grounding required for functional segmentation. To address this, we curated a high-fidelity synchronized corpus by intersecting DALI v2.0 and the Harmonix Set.

Although the resulting dataset comprises 148 tracks, it represents a high-density ground truth where every lyrical line is precisely anchored to both a temporal window and a structural function. This 148-song subset yields a total of $N = 9662$ individual lyric lines, providing a substantial sample size for fine-grained supervised learning. Unlike massive but unaligned text scrapings, this synchronized multi-modal corpus enables the model to learn the exact transition points between musical sections.

The synchronization is performed in two rigorous steps:

1. **Metadata Matching:** Identify overlapping tracks between the two datasets using a combination of fuzzy string matching for titles and artists, followed by manual verification of unique track IDs to ensure data integrity.

2. **Timestamp-based Label Fusion:** Assign a structural label to each lyric line by aligning its temporal position with the annotated section boundaries. A line L_i is directly mapped and assigned the label y_i of the active structural segment corresponding to the line's onset time.

By utilizing this synchronized multi-modal corpus, our study prioritizes annotation fidelity over raw volume. This ensures that the structural boundaries learned by the model are physically grounded in the audio-textual timeline, providing a more reliable foundation for Music Structure Analysis (MSA) than orders-of-magnitude larger ungrounded datasets.

Table 3.1: Features of the synchronized dataset constructed from the intersection of DALI v2.0 and the Harmonix Set.

Data Feature	**Description**
Audio File	Original music recording for acoustic analysis and temporal alignment.
Mel-Spectrogram	Time–frequency representation generated from the audio signal for compatibility with audio-based MSA models.
Start Time	Starting timestamp of each lyric line obtained from DALI annotations.
End Time	Ending timestamp of each lyric line obtained from DALI annotations.
Line-Level Lyrics	Lyric lines synchronized with the corresponding temporal interval.
Segment Label	Structural annotation inherited from the Harmonix Set, including Intro, Verse, Pre-Chorus, Chorus, Bridge, Outro, and Instruments.
Song Title	Matched song title shared by both DALI v2.0 and Harmonix Set.
Artist	Performer or creator of the song.

3.1.1 Labeling and Class Distribution

To ensure comprehensive structural modeling, the classifier's label space $|\mathcal{C}| = 7$ is defined over seven classes. This set includes six primary functional categories—*Intro, Verse, Pre-Chorus, Chorus, Bridge, and Outro*—and an additional *Instruments* category for non-lyrical interludes and transitions.

The class distribution across the 148-song synchronized dataset (totaling $N = 9662$ lines) is summarized in Table 3.2. The corpus is predominantly composed of *Chorus* (48.45%) and *Verse* (30.00%) segments, which together constitute nearly 80% of the data. This distribution highlights the inherent data imbalance in music structure, particularly for transitional sections like *Outro* (1.08%), providing a realistic benchmark for automated classification.

While the label space $\mathcal{C}$ includes an *Instruments* category to maintain sequence continuity, its support in the current synchronized textual corpus is zero. This is because our dataset construction is based on line-level lyrics; purely instrumental interludes that lack corresponding text were naturally excluded during the text-to-label mapping process. However, this category is strategically retained in the classifier's output layer to maintain architectural compatibility with state-of-the-art audio-based benchmarks and to facilitate the future integration of multimodal features for silent segment detection. To ensure the integrity of the evaluation, this category is excluded from the calculation of aggregate performance metrics, such as the weighted F1-score, as it does not contribute to the linguistic variance of the current lyric-based modeling task.

3.1.2 Sequence Representation and Input Preparation

The synchronized data are stored in a structured format where each song is represented as a sequence of lyric lines $\mathcal{L} = \{L_1, L_2, \ldots, L_N\}$. For each line L_i, we derive four feature components:

- A local context window (W_i)
- A sectional context (S_i)

Table 3.2: Class Distribution of The Synchronized Lyric Dataset

Category	**Support (Lines)**	**Percentage (%)**
Chorus	4681	48.45%
Verse	2899	30.00%
Bridge	1027	10.63%
Pre-Chorus	745	7.71%
Intro	206	2.13%
Outro	104	1.08%
Instruments	0	0.00%
Total	**9662**	**100.00%**

- A normalized position (P_i)
- A repetition feature (R_i)

Chapter 4 System Architecture and Methodology

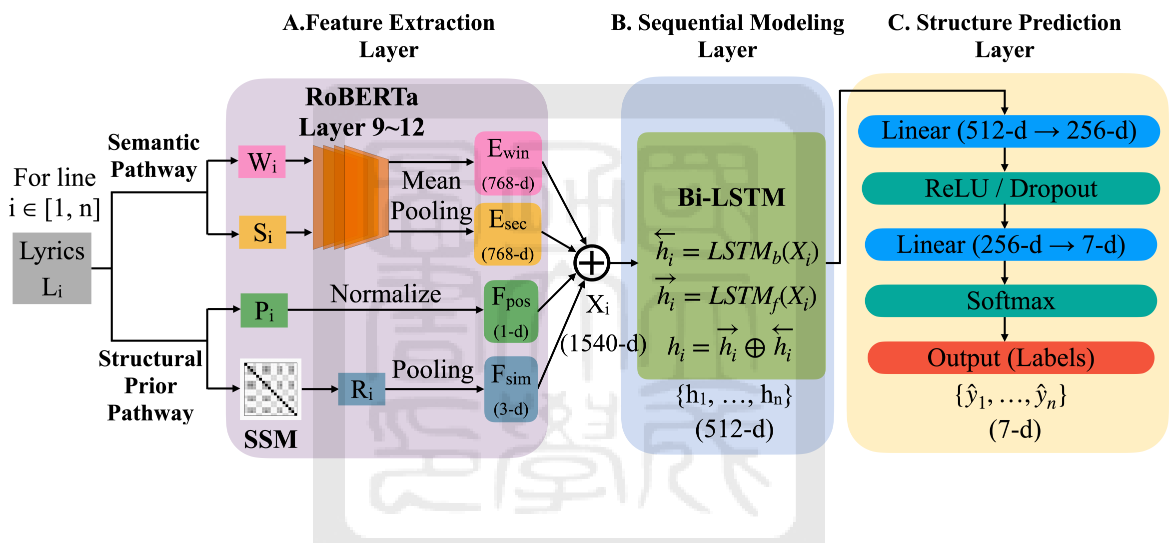

Figure 4.1: Overall framework for lyrics-based music structure analysis.

We formulate music structure analysis as a sequence labeling task over individual lyric lines. Given a song represented as a sequence of lines:

$$\mathcal{L} = \{L_1, L_2, \ldots, L_N\}, \tag{4.1}$$

the objective is to predict a corresponding sequence of structural labels:

$$\mathcal{Y} = \{y_1, y_2, \ldots, y_N\}, \quad y_i \in \mathcal{C}, \tag{4.2}$$

where $\mathcal{C}$ denotes the predefined set of structural categories (e.g., Verse, Chorus).

Each line L_i is projected into a high-dimensional feature vector $X_i \in \mathbb{R}^d$, and the model learns a mapping function:

$$f : (X_1, \dots, X_N) \rightarrow (y_1, \dots, y_N). \tag{4.3}$$

This formulation enables the joint modeling of local semantic content and global structural dependencies.

4.1 Feature Extraction

As illustrated in Fig. 4.1, each lyric line L_i is encoded into four complementary feature representations to capture diverse textual and contextual attributes.

Semantic Features

To capture both local and paragraph-level semantics, we utilize a pre-trained RoBERTa encoder. The window context $W_i = \{L_{i-2}, L_{i-1}, L_i, L_{i+1}, L_{i+2}\}$ and the section context $S_i = \{L_j \mid L_j \in \text{Paragraph}(L_i)\}$ are encoded into 768-dimensional embeddings respectively as:

$$E_{win} = f_{\text{RoBERTa}}(W_i), E_{sec} = f_{\text{RoBERTa}}(S_i). \tag{4.4}$$

Paragraph boundaries are determined solely by double line breaks within the raw text files, independent of any temporal or structural annotations.

Positional Features

To provide the model with a sense of temporal progression, the relative position of each line is defined as:

$$P_i = \frac{i}{N}, \tag{4.5}$$

which is further normalized into a scalar feature F_{pos}.

Repetition Features: Explicit Repetition Awareness

Repetition is a foundational principle of musical form, serving as a primary cue for identifying functional anchors such as the *Chorus*. To bridge the gap between acoustic analysis and lyrical processing, we introduce **Explicit Repetition Awareness** through the feature $F_{sim} \in \mathbb{R}^3$. This design represents a textual transition of the Self-Similarity Matrix (SSM), a cornerstone technique in audio-based MSA. While general-purpose NLP models rely on implicit attention to link recurring tokens, they often struggle to perceive the "piecewise functional" nature of music—where long-range textual identity signifies a formal shift in structural state. By injecting F_{sim}, we explicitly inform the model of the semantic recurrence density, effectively providing a low-dimensional proxy for high-dimensional acoustic similarity patterns.

For each song, we compute a pairwise semantic similarity matrix $S \in \mathbb{R}^{N \times N}$ based on the cosine similarity of window embeddings. To prevent the model from focusing on trivial self-matches, the diagonal elements are penalized as follows:

$$S_{i,j} = \begin{cases} \frac{\mathbf{e}_i \cdot \mathbf{e}_j}{\|\mathbf{e}_i\| \|\mathbf{e}_j\|}, & \text{if } i \neq j. \\ -1, & \text{if } i = j. \end{cases} \tag{4.6}$$

where $\mathbf{e}_i$ and $\mathbf{e}_j$ represent the contextual embeddings of the i-th and j-th lyric lines, respectively. For each line L_i, we extract a three-dimensional repetition vector $R_i = [r_i^{(1)}, r_i^{(2)}, r_i^{(3)}]$ from the similarity set $\mathcal{S}_i = \{S_{ij} \mid j \neq i\}$:

- $r_i^{(1)} = \max(\mathcal{S}_i)$: The maximum similarity with any other line, identifying the most probable recurring counterpart.
- $r_i^{(2)} = \mathbb{I}(r_i^{(1)} > \tau_1)$: A binary indicator for strong semantic identity, marking potential structural recursion.
- $r_i^{(3)} = \sum_{s \in \mathcal{S}_i} \mathbb{I}(s > \tau_2)$: The frequency count of lines exceeding a similarity threshold, capturing the thematic density of the section.

where the thresholds $\tau_1 = 0.85$ and $\tau_2 = 0.8$ were determined through systematic hyperparameter optimization.

To ensure stability across songs with varying lyrical densities, the use of cosine similarity provides a normalized scale within $[-1, 1]$, allowing these thresholds to function as robust indicators of semantic recurrence regardless of the original embedding magnitudes. Sensitivity analysis indicates that the model performance remains stable within a $\pm 10\%$ range of these values. While learned thresholds were considered, we opted for fixed optimized parameters to minimize computational graph complexity, maintaining the model's efficiency for resource-constrained environments. The resulting vector R_i is utilized as the repetition feature $F_{sim} \in \mathbb{R}^3$, acting as a **musical knowledge injection** that empowers the Bi-LSTM to distinguish between narrative-driven low-repetition segments (e.g., Verse) and high-density thematic recurrences (e.g., Chorus).

Complexity and Inference Scalability: Quantifying the computational overhead is essential for edge deployment. Although generating the similarity matrix $S \in \mathbb{R}^{N \times N}$ involves a complexity of $O(N^2)$, the practical burden remains minimal. In typical popular music, the number of lyric lines N is small (usually $N < 100$), leading to fewer than 10^4 operations per song. In contrast, audio-based models must process high-resolution spectrograms with sequence lengths L often exceeding 10^4 to 10^5, where $O(L^2)$ attention mechanisms become prohibitive for edge devices.

Furthermore, while the full RoBERTa-base encoder (124.65 M parameters) is used in this study, the framework's modularity allows for the seamless integration of lightweight distilled encoders (e.g., DistilRoBERTa or TinyBERT) to further reduce inference latency and memory footprint. By shifting the structural analysis from the high-bandwidth audio domain to the low-dimensional semantic domain, the proposed approach achieves an improved balance between predictive power and hardware feasibility in Multimedia IoT scenarios.

Feature Integration Strategy

The concatenation of F_{pos} and F_{sim} with deep RoBERTa embeddings incorporates domain-specific musical knowledge into the framework. Unlike general natural language processing, where semantic flow is primary, MSA relies heavily on *quasi-periodicity* and *temporal templates.*

General-purpose models like Transformers often suffer from "attention dilution," where structural regularities are lost in complex global attention weights. By explicitly injecting F_{pos}, we provide a structural pointer that constrains the label space based on the song's progression (e.g., distinguishing an early Verse from a late Outro). Similarly, F_{sim} transforms the latent task of pattern recognition into an explicit structural anchor, mirroring the Self-Similarity Matrix (SSM) approach used in acoustic analysis. This ensures the Bi-LSTM prioritizes structural rhythm over purely sequential linguistic features.

Feature Fusion

The final multi-modal representation X_i is formed by concatenating all features:

$$X_i = E_{win} \oplus E_{sec} \oplus F_{pos} \oplus F_{sim}, \tag{4.7}$$

resulting in a 1540-dimensional vector.

4.2 Temporal Modeling

The sequence $\{X_1, \ldots, X_N\}$ is processed by a bidirectional Long Short-Term Memory (Bi-LSTM) network to capture long-range dependencies:

$$\overrightarrow{h_i} = \mathrm{LSTM}_f(X_i), \quad \overleftarrow{h_i} = \mathrm{LSTM}_b(X_i). \tag{4.8}$$

The contextual representation h_i is obtained via concatenation:

$$h_i = \overrightarrow{h_i} \oplus \overleftarrow{h_i}, \tag{4.9}$$

yielding a 512-dimensional hidden state.

4.3 Classification

The hidden representation h_i is passed through a feed-forward classifier with ReLU activation and dropout for regularization:

$$z_i = \mathrm{ReLU}(\mathrm{Linear}_{512 \to 256}(h_i)). \tag{4.10}$$

Finally, a Softmax layer projects the vector into the label space:

$$\hat{y}_i = \mathrm{Softmax}\left(\mathrm{Linear}_{256 \to |\mathcal{C}|}(z_i)\right), \tag{4.11}$$

where $\hat{y}_i$ denotes the predicted probability distribution over $|\mathcal{C}| = 7$ structural categories for line L_i. As established in Section 3.2, this label space accounts for both the six primary functional sections and the additional *Instruments* category to ensure sequence continuity. By mapping high-dimensional feature fusion into this specific categorical distribution, the classifier effectively handles the class imbalances reported in Table 3.2, providing robust structural predictions across both dominant (e.g., *Chorus*) and minority (e.g., *Outro*) segments.

Chapter 5 Experimental Results and Analysis

5.1 Experimental Setup

1. **Evaluation Protocol and Metric Rationale:** The proposed model was evaluated using 5-fold cross-validation with a fixed random seed = 43 to ensure statistical robustness. Our folds are split by song, ensuring that no lines from the same song appear in both training and testing sets simultaneously. Unlike traditional audio-based Music Structure Analysis (MSA) that focuses solely on temporal boundaries, our evaluation employs a multi-dimensional framework to assess both classification accuracy and boundary precision.

   By leveraging the precise line-level timestamps from the DALI v2.0 dataset, we conduct evaluation across three primary dimensions:

   - **Classification Performance:** We report the precision, recall, and F1-score for each structural category (e.g., Verse, Chorus), accompanied by their respective means and standard deviations. To provide a holistic performance summary, the **Weighted F1-score** is utilized as the primary aggregate metric to account for class imbalance across musical sections.
   - **Line-level Accuracy:** To evaluate the overall labeling precision, we report the **Accuracy** measured at the line-level. This metric quantifies the proportion of individual lyric lines for which the predicted structural label matches the ground truth. It reflects the model's reliability in identifying the semantic state of each textual segment within the song's progression.

- **Structural Segmentation:** We employ the **L-Measure** to assess the global structural integrity. Although the model operates on discrete lyrics lines, each prediction is mapped back to the temporal domain using the DALI v2.0 timestamps. The L-Measure evaluates whether pairs of time points throughout the song are correctly assigned to the same or different semantic categories. This captures the model's ability to maintain sequential consistency and accurately reconstruct the organizational "skeleton" of the musical work.

This comprehensive metric suite is particularly relevant for Multimedia IoT applications, such as real-time lyric indexing and automated music summarization. Notably, as the *Instruments* category contains zero samples in this lyrics-based dataset, it was excluded from metric calculations to prevent skewed results. However, the architectural output for *Instruments* was retained to ensure compatibility for future cross-modal integration with audio features. This setup provides a rigorous assessment of the model's capacity to recognize both dominant sections (e.g., Chorus, Verse) and transitional segments (e.g., Intro, Outro).

2. **Implementation Details:** The model was implemented in PyTorch and trained on an NVIDIA A100 GPU. Hyper-parameters were determined through systematic tuning using Optuna and summarized as follows.

   - Learning Rate: 1.88×10^{-4}
   - Weight Decay: 1.104×10^{-3}
   - Dropout Rate: 0.1
   - Optimizer: AdamW
   - Loss Function: The model was optimized using the cross-entropy loss function, which measures the discrepancy between the predicted probability distribution and the ground-truth one-hot encoded labels. This objective function is formulated as:

$$\mathcal{L} = -\sum_{t=1}^{N}\sum_{c=1}^{C} y_{t,c}\log(\hat{y}_{t,c}), \tag{5.1}$$

where N denotes the sequence length, C is the number of structural classes, $y_{t,c}$ is the binary indicator of whether the t-th lyric line belongs to class c, and $\hat{y}_{t,c}$ denotes the predicted probability that the t-th lyric line belongs to class c.

5.2 Feature Strategy

5.2.1 Analysis of RoBERTa Layer Selection

To determine the optimal semantic representation from the pre-trained RoBERTa-base model, we conducted a comparative analysis of different hidden-state extraction strategies. While the final layer (Layer 12) often captures fine-grained syntactic and lexical features, structural segmentation in music relies on a higher level of contextual consistency across lyrical segments. We hypothesize that a hierarchical semantic extraction approach—specifically averaging the last four layers—is more aligned with the perceptual hierarchy of Music Structure Analysis (MSA). In transformer architectures, mid-to-high layers are documented to encode richer paragraph-level context, which is essential for identifying the thematic unity of sections like the *Chorus* or the narrative arc of the *Verse*.

Task Definition for MAP: To validate this hypothesis, we defined a *Contextual Retrieval Task* using Mean Average Precision (MAP) to measure the model's ability to align local semantic content with higher-level structural context. For each lyric line L_i in a song, let E_{win} be the window embedding and E_{sec} be the embedding of its ground-truth section. We calculate the cosine similzero-cost semantic optimizationarity between E_{win} and a set of candidate section embeddings $\mathcal{S} = \{E_{sec}^{(1)}, E_{sec}^{(2)}, ..., E_{sec}^{(m)}\}$, where m is the number of structural sections in the song. The Average Precision (AP) for each query is defined as:

$$AP = \frac{\sum_{k=1}^{m}(P(k) \times rel(k))}{\text{Total relevant sections}}, \tag{5.2}$$

where $P(k)$ is the precision at rank k, and $rel(k)$ is a binary indicator of whether the k-th ranked section is the ground-truth section. The *Mean Average Precision* (MAP) is then computed as the mean of AP values across all N lyric lines in the dataset:

$$MAP = \frac{1}{N}\sum_{i=1}^{N} AP_i. \tag{5.3}$$

This task ensures that the selected strategy effectively captures the hierarchical semantic coherence necessary for distinguishing functional musical blocks.

Refinement Analysis: As summarized in Table 5.1, four strategies were evaluated. Although the numerical gain in MAP for *Last 4 Average* (0.4994) over *Last 4 Concat* (0.4993) is marginal, the former is strategically improved for resource-efficient modeling. By providing a compact 768-dimensional representation—compared to the 3072-dimensional vector of the concatenation strategy—the averaging approach significantly reduces the parameter count and computational complexity of the subsequent Bi-LSTM layers.

Furthermore, since any hidden-state extraction necessitates a complete 12-layer forward pass, utilizing the averaged representation from multiple high-level layers represents a zero-cost semantic optimization. This ensures the highest fidelity semantic input for structural transition modeling without incurring additional inference latency.

Table 5.1: RoBERTa Layer Selection Strategies and Performance

Strategy	Description	Dim.	MAP
Last Only	Output of Layer 12	768	0.4973
Second-to-Last	Output of Layer 11	768	0.4985
Last 4 Concat.	Concat. of Layer 9–12	3072	0.4993
Last 4 Average	**Mean of Layer 9–12**	**768**	**0.4994**

Computational Complexity Analysis: Detailed profiling reveals that the pre-trained RoBERTa-base encoder (124.65 M parameters) occupies 475.49 MB of static GPU memory. It is critical to note that the *Last 4 Average* strategy incurs no additional memory overhead compared to the *Last Only* approach during inference. To maximize efficiency in resource-constrained environments, we employ a feature caching strategy. By pre-computing these 1540-dimensional fused features and storing them offline, we decouple the heavy transformer inference from the training phase. This implementation allows the peak GPU memory during the 5-fold cross-validation of the Bi-LSTM to be optimized at 189.3 MB, achieving an improved balance between **perceptual depth** and **deployment scalability**.

5.2.2 Global Information Modeling Method Selection

To assess the capability of different architectural backbones in modeling global structural dependencies, we compared the proposed Bi-LSTM with two representative alternatives: a *Cascading Model* and a *Hybrid Model*.

The *Cascading Model* adopts a sequential integration strategy where diverse feature sets—including RoBERTa-based window and section embeddings, normalized positional features, and statistical representative features from the similarity matrix—are first concatenated into a unified 1540-dimensional vector. This concatenated representation is then projected and fed into a Transformer Encoder to capture long-range dependencies across the entire sequence before final classification.

In contrast, the *Hybrid Model* employs a parallelized feature processing approach. Specifically, only the window embeddings are processed through a Transformer Encoder to extract latent sequential patterns, while other features (section embeddings, positional, and representative features) bypass the Transformer. These context-aware latent representations are subsequently fused with the remaining raw features at a final concatenation layer to produce the class probabilities.

As reported in Table 5.2, the Bi-LSTM achieves consistently strong performance across structural categories, reaching a weighted F1-score of 0.7456. It should be explicitly noted that the results presented in this preliminary architecture comparison represent the baseline performance using default hyperparameters and a fixed data split. This initial screening was conducted specifically to identify the most robust backbone before proceeding to fine-grained optimization.

In contrast, the final proposed model, which incorporates systematic hyperparameter optimization using Optuna and 5-fold cross-validation, achieves an improved weighted F1-score of 0.7848. This performance discrepancy (0.7456 vs. 0.7848) reflects the gains attributed to the fine-tuning of hidden dimensions, dropout rates, and learning schedules, as further detailed in the ablation analysis in Section 5.4.

Table 5.2: Global Information Modeling Method Selection (Weighted F1-score before hyperparameter optimization)

Model Architecture	Weighted F1-score
Cascading Model	0.7042
Hybrid Model	0.7041
Proposed Bi-LSTM	**0.7456**

5.3 Model Comparison

5.3.1 Comparison with Lyrics-Only Baselines

We compared the proposed framework with two significant baselines to evaluate its effectiveness: **LongFormer**, a Transformer-based model optimized for long-range textual dependencies, and **SongFormer**, the current state-of-the-art model in music sequence modeling. While SongFormer primarily operates on acoustic data, its inclusion here (as summarized in Table 5.3) serves as a cross-modal benchmark to demonstrate the competitive performance of a lyrics-only approach against established audio-driven methods.

Experimental Setup and Baseline Parity: To ensure a rigorous benchmark, we employed LongFormer (*longformer-base-4096*) as the primary textual baseline, configured with a maximum sequence length of 1,024 tokens to accommodate comprehensive lyrical context. While LongFormer relies on sliding-window and global attention mechanisms to process long sequences, our proposed Bi-LSTM adopts a feature-augmented approach. By explicitly integrating domain-specific musical knowledge—such as structural repetition patterns and relative positional encodings—our model demonstrates that a parameter-efficient architecture, when guided by music-aware features, can surpass the structural disambiguation capabilities of significantly larger, generic Transformer-based models.

Overall Performance Analysis: As summarized in Table 5.3, the proposed model consistently outperforms the LongFormer baseline across all evaluation metrics. Specifically, our model achieved a Weighted F1-score of 0.7848, representing an improvement of 21.76% over LongFormer (0.5672).

A similar trend is observed in Accuracy and L-Measure, where the proposed model reached 0.7867 and 0.7948 respectively, substantially surpassing LongFormer's scores of 0.6142 and 0.6398. Furthermore, the proposed model exhibits a lower standard deviation in Weighted F1 and Accuracy, suggesting not only improved performance but also enhanced training stability compared to the LongFormer architecture.

Table 5.3: Overall Performance Comparison of Different Models (Mean ± Std)

Model	Weighted F1	Accuracy	L-Measure
Proposed	**0.7848 ± 0.0178**	**0.7867 ± 0.0175**	**0.7948 ± 0.0317**
LongFormer	0.5672 ± 0.0408	0.6142 ± 0.0397	0.6398 ± 0.0432
SongFormer	0.5254 ± 0.0251	0.5448 ± 0.0193	0.6283 ± 0.0087

Sectional Performance Discussion: Detailed sectional F1-scores are illustrated in Fig. 5.1. The analysis reveals several key insights based on the comparison between our proposed model (Table 5.4) and the LongFormer baseline (Table 5.5).

Analysis of dominant sections shows that for *Chorus* and *Verse*, our proposed model achieves high F1-scores of 0.8782 and 0.7830, respectively. While the LongFormer model shows some capability in these frequent sections (Chorus: 0.7232, Verse: 0.6079), it struggles to match the precision and consistency of our feature-augmented Bi-LSTM approach, which benefits from specialized musical feature integration.

A notable difference occurs in transitional and boundary sections such as the *Intro* and *Outro*. For the *Intro*, our model achieves an F1-score of 0.5798, significantly outperforming the LongFormer's 0.1038. This trend is even more pronounced in the *Outro*, where the LongFormer model failed completely (F1: 0.0000), whereas our model maintains a functional F1-score of 0.4429. Furthermore, our model handles the *Pre-*

Chorus much more effectively (F1: 0.6210 vs. 0.0578), suggesting that the proposed features better capture the preparatory harmonic shifts preceding the chorus.

Even in the most challenging category, the *Bridge*, our model (F1: 0.5649) significantly outperforms the baseline (F1: 0.2477), demonstrating that our architecture is better suited for capturing the non-linear structural transitions inherent in popular music.

Computational and Resource Efficiency: The proposed framework achieves massive efficiency gains over the Transformer-based baseline. As illustrated in Fig. 5.2, our model utilizes only (189.3 MB of peak VRAM, representing a significant 42.4× reduction compared to LongFormer’ s 8,028.2 MB. Furthermore, the training speed is significantly enhanced, with our model completing an epoch in 1.9 minutes—a 78.1× speedup over LongFormer’s 150.0 minutes. This drastic reduction in both memory footprint and temporal cost highlights the efficiency of our lyrics-driven approach, making it exceptionally well-suited for Multimedia IoT environments and real-world applications where computational resources are limited.

Discussion on Architectural Efficiency: The significant performance gap between the proposed model and LongFormer (Weighted F1: 0.7848 vs. 0.5672) validates the effectiveness of the inductive bias strategy. LongFormer attempts to ”discover” structural patterns from raw sequences through self-attention, which requires vast amounts of data and computational power to converge.

In contrast, our framework bypasses the need for the model to learn the concept of ”repetition” from scratch. By supplying F_{sim} as a pre-computed signal, the Bi-LSTM can focus its capacity on learning the *functional mapping* of these repetitions to musical sections. This theoretical shift allows a much lighter architecture to achieve improved structural disambiguation, particularly in transitional segments like the Pre-Chorus and Bridge where LongFormer’s attention mechanism tends to be overly diffuse.

5.3.2 Comparison with Audio-Only Baselines

To further evaluate cross-modal effectiveness, we compared the proposed lyrics-driven model with the audio-based **SongFormer** to analyze the inherent trade-offs between

Table 5.4: Sectional Performance Metrics for Proposed Model (Mean ± Std) with Class Support

Section	**Precision**	**Recall**	**F1-Score**	**Support**	**Percentage**
Intro	0.8152 ± 0.1062	0.4539 ± 0.0485	0.5798 ± 0.0497	206	2.13%
Verse	0.8337 ± 0.0427	0.7424 ± 0.0563	0.7830 ± 0.0300	2899	30.00%
Pre-Chorus	0.5863 ± 0.0415	0.6671 ± 0.0570	0.6210 ± 0.0183	745	7.71%
Chorus	0.8585 ± 0.0175	0.8995 ± 0.0200	0.8782 ± 0.0108	4681	48.45%
Bridge	0.5546 ± 0.1084	0.6194 ± 0.1606	0.5649 ± 0.0796	1027	10.63%
Outro	0.6583 ± 0.2585	0.4028 ± 0.2308	0.4429 ± 0.2492	104	1.08%
Total	–	–	–	**9662**	**100.00%**

Table 5.5: Sectional Performance Metrics for LongFormer Model (Mean ± Std) with Class Support

Section	**Precision**	**Recall**	**F1-Score**	**Support**	**Percentage**
Intro	0.4500 ± 0.3930	0.0606 ± 0.0562	0.1038 ± 0.0923	206	2.13%
Verse	0.6180 ± 0.0644	0.6029 ± 0.0992	0.6079 ± 0.0737	2899	30.00%
Pre-Chorus	0.1241 ± 0.1535	0.0383 ± 0.0479	0.0578 ± 0.0713	745	7.71%
Chorus	0.6338 ± 0.0470	0.8455 ± 0.0497	0.7232 ± 0.0386	4681	48.45%
Bridge	0.4852 ± 0.1778	0.1850 ± 0.0815	0.2477 ± 0.0920	1027	10.63%
Outro	0.0000 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000	104	1.08%
Total	–	–	–	**9662**	**100.00%**

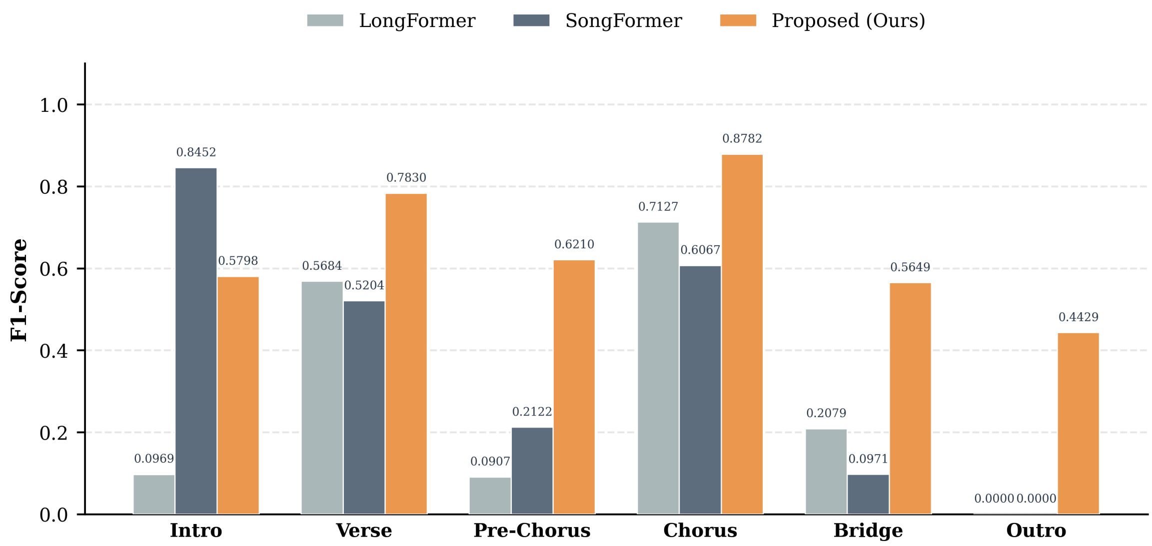

Figure 5.1: Performance comparison with lyrics-only and audio-only baselines (F1-score).

acoustic-driven and semantic-driven approaches.

SongFormer Training and Alignment Protocol: To ensure a rigorous cross-modal comparison, SongFormer was retrained from scratch on our synchronized dataset. We adopted the original architecture and optimized hyperparameters, implementing an identical 5-fold cross-validation setup. Since SongFormer generates structural predictions at fixed temporal intervals (audio-frame-based), while our model operates on discrete lyric lines (line-based), a temporal-to-textual mapping was performed. We utilized precise timestamps for each lyric line; a structural label was assigned to a line if the majority of SongFormer's temporal predictions within that line's duration matched that label. This synchronization ensures a direct, line-by-line performance comparison.

Overall Performance Analysis: As illustrated in Table 5.3, our proposed model demonstrates a consistent advantage over the SongFormer baseline across all key metrics, although part of this gain may come from the explicit structural features rather than the backbone architecture itself. Notably, the proposed model achieves a Weighted F1-score of 0.7848, which represents a substantial improvement over SongFormer's 0.5254.

This performance gap extends to classification precision and structural boundary detection. In terms of Accuracy, our model (0.7867) demonstrates competitive performance compared with SongFormer (0.5448) by 44.40%, suggesting that the integration of specialized musical features significantly mitigates the misclassification issues prevalent in the SongFormer architecture. Furthermore, the L-Measure—a critical metric for structural boundary consistency—shows our model reaching 0.7948 compared to SongFormer's 0.6283.

Comparative Analysis of Lyrics and Audio Modalities: The proposed model demonstrates stronger performance in identifying core functional sections, which may be attributed to the semantic characteristics of lyrics. Audio signals are inherently high-bandwidth and non-stationary; during structural transitions, changes in instrumentation, timbre, and dynamics can introduce acoustic variability that makes structural boundaries more difficult to distinguish. In contrast, lyrics provide a compact semantic representation that is less affected by performance-related variations. Repeated lyrical phrases in the *Chorus* and the narrative progression of the *Verse* offer consistent semantic cues that facilitate the identification of these functional sections. Consequently, the proposed Bi-LSTM framework can more effectively capture structural patterns from lyrics, whereas audio-based models may experience greater ambiguity when acoustically similar sections exhibit overlapping spectral characteristics.

Performance Analysis and Modality Trade-offs: Detailed sectional performance comparison between the proposed model and SongFormer is illustrated in Fig. 5.1, Table 5.4, and Table 5.6. Key observations include:

Our model achieves substantially higher F1-scores for *Verse* (0.7830 vs. 0.5204) and *Chorus* (0.8782 vs. 0.6067) than SongFormer. These results suggest that lyrical information, particularly repeated textual patterns, provides strong structural cues for identifying these major functional sections. By explicitly incorporating repetition-aware features, the proposed framework is better able to exploit these cues, whereas audio-based models may encounter greater ambiguity when relying solely on acoustic patterns.

Furthermore, SongFormer exhibits an exceptional F1-score for the *Intro* (0.8452), outperforming our model (0.5798). This suggests that the initial structural onset often possesses distinct acoustic energy patterns that SongFormer identifies more readily. However, this capability does not generalize to the end of the track; for the *Outro*, SongFormer fails completely (F1: 0.0000), whereas our model maintains a functional F1-score of 0.4429 by leveraging textual structural cues that persist even as acoustic signals diminish.

In complex segments such as the *Pre-Chorus* and *Bridge*, our proposed approach (0.6210 and 0.5649, respectively) demonstrates competitive performance compared with SongFormer (0.2122 and 0.0971). The low performance of the baseline in these categories suggests that audio models struggle with the high variance of transitional musical passages, whereas our low-bandwidth semantic proxy provides a much more stable reference for identifying structural shifts.

Table 5.6: Sectional Performance Metrics for SongFormer Model (Mean ± Std) with Class Support

Section	**Precision**	**Recall**	**F1-Score**	**Support**	**Percentage**
Intro	0.8253 ± 0.1658	0.9054 ± 0.0968	0.8452 ± 0.0868	206	2.13%
Verse	0.4829 ± 0.0413	0.5792 ± 0.0784	0.5204 ± 0.0152	2899	30.00%
Pre-Chorus	0.3232 ± 0.1219	0.2240 ± 0.2123	0.2122 ± 0.1465	745	7.71%
Chorus	0.6160 ± 0.0448	0.6018 ± 0.0431	0.6067 ± 0.0244	4681	48.45%
Bridge	0.1036 ± 0.0948	0.0947 ± 0.0906	0.0971 ± 0.0908	1027	10.63%
Outro	0.0000 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000	104	1.08%
Total	–	–	–	**9662**	**100.00%**

Computational and Resource Efficiency: The proposed framework achieves considerable efficiency gains over the audio-based baseline. As shown in Fig. 5.2, our model utilizes only 189.3 MB of peak VRAM, representing a 20.5× reduction compared to SongFormer's 3890 MB. This validates that utilizing a semantic proxy is not only more accurate for core sections but also far more suitable for resource-constrained Multimedia IoT environments where memory and power are critical constraints.

Discussion on Model Generalization: While high-capacity audio models like Song-

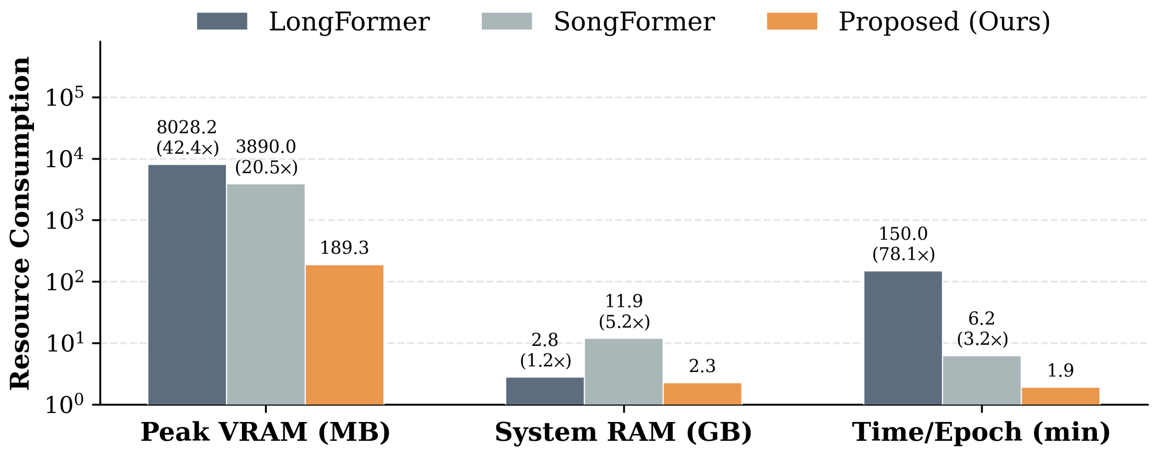

Figure 5.2: Resource Consumption Analysis: Comparison of peak GPU memory usage between the proposed model, SongFormer, and LongFormer baselines.

Former typically benefit from larger training corpora, our results demonstrate a critical trade-off: in data-constrained environments, the low-dimensional and semantic-rich nature of lyrics allows a lightweight Bi-LSTM to achieve faster convergence and more reliable structural labeling. This confirms the effectiveness of the lyrics modality as a primary source for real-time music understanding tasks.

5.3.3 Detailed Error Analysis via Confusion Matrix

To further investigate the classification behavior of the proposed framework beyond aggregate evaluation metrics, Fig. 5.3 presents the normalized confusion matrix averaged over the five-fold cross-validation. The matrix provides detailed insights into the strengths and remaining limitations of the proposed music-aware representation.

Overall, the proposed framework demonstrates strong discriminative capability for the major functional sections of popular music. Among all categories, *Chorus* achieves the highest recognition rate (89.96%), indicating that repetitive lyrical patterns provide highly distinctive semantic cues for chorus identification. Likewise, *Verse* is recognized with a relatively high accuracy (74.51%), suggesting that the hierarchical semantic representation effectively captures the narrative progression that characterizes verse sections. The *Pre-Chorus* also achieves satisfactory recognition (65.77%), demonstrat-

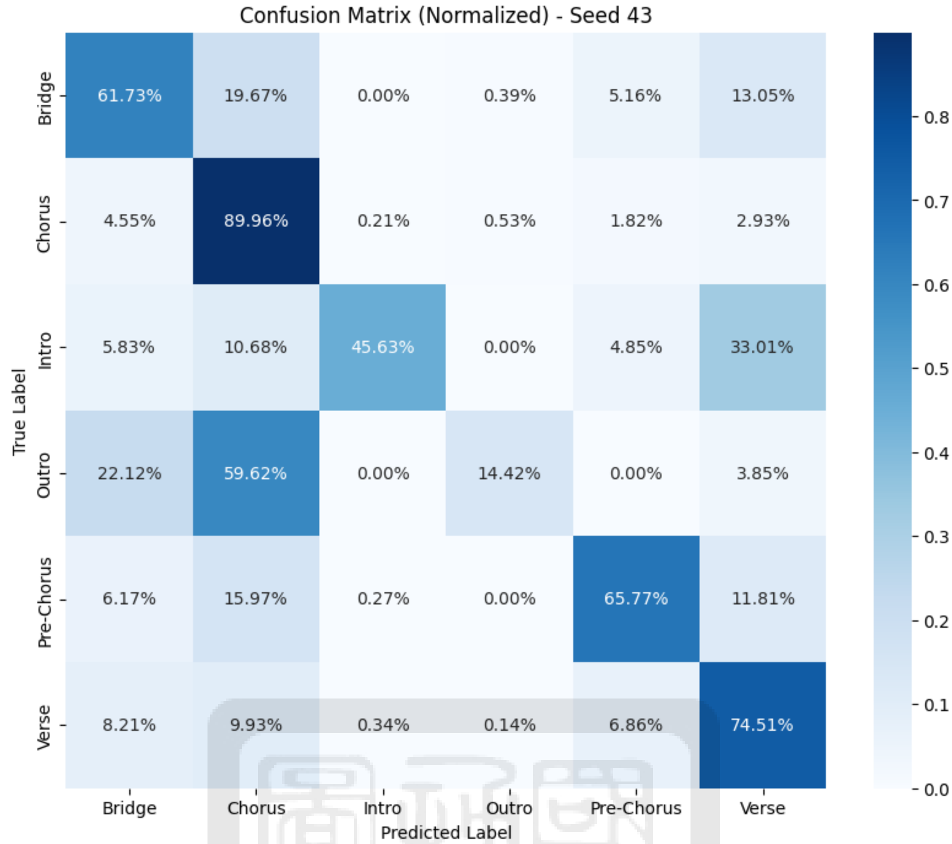

Figure 5.3: Normalized confusion matrix of the proposed framework averaged over five-fold cross-validation.

ing that the incorporation of repetition-aware and positional features improves the identification of transitional sections that are typically difficult for general-purpose language models.

Most classification errors occur between structurally adjacent sections rather than semantically unrelated ones. For example, approximately 33.01% of *Intro* samples are classified as *Verse*. This confusion is expected because both sections often contain unique, non-repetitive lyrical content and lack the strong repetitive characteristics commonly associated with choruses. Similarly, *Bridge* is occasionally misclassified as either *Chorus* (19.67%) or *Verse* (13.05%), reflecting the high stylistic diversity and relatively weak lexical regularity of bridge sections in popular music.

The most challenging category remains *Outro*. As shown in Fig. 5.3, 59.62% of Outro samples are predicted as *Chorus*. This observation is consistent with common song-writing practice, where many popular songs conclude by repeating the final chorus. Although the lyrical content is nearly identical, the functional role differs, making it difficult to distinguish these sections using textual information alone. This limitation highlights an inherent challenge of lyrics-only music structure analysis and suggests that lightweight acoustic cues may further improve discrimination between repeated choruses and song endings.

Overall, the confusion patterns closely align with the compositional characteristics of popular music. Rather than exhibiting arbitrary misclassifications, the proposed framework primarily confuses structurally related sections, indicating that the learned representations successfully capture meaningful musical organization. These findings further support the effectiveness of explicitly incorporating music-aware structural priors, including hierarchical semantic representations, repetition-aware similarity, and positional information, for lyrics-based music structure analysis.

5.4 Ablation Analysis

To quantify the contribution of each component and validate our theoretical framework, we conducted a systematic ablation study using the weighted F1-score as the primary evaluation metric. Table 5.7 summarizes the performance of various configurations and the corresponding degradation relative to the full model.

The results highlight that the integration of *Section-level embeddings* (E_{sec}) is the most critical factor for capturing the global structural coherence of lyrics. When removing E_{sec} alongside the structural features (F_{pos}, F_{sim}), the weighted F1-score drops significantly by 13.57% (from 0.7848 to 0.6783). This underscores the necessity of higher-level semantic representations; without sectional context, the model struggles to disambiguate lines that are semantically similar but belong to different functional blocks.

In contrast, excluding only the *Repetition* and *Positional* features (F_{sim}, F_{pos}) leads to a marginal numerical decrease of 0.55%. However, from a structural modeling perspective, these features serve as a vital lyrical inductive bias. While the Bi-LSTM can partially infer patterns from deep embeddings, F_{pos} and F_{sim} act as a structural skeleton that explicitly guides the model. F_{pos} functions as a temporal anchor that constrains the label space based on the song's progression, while F_{sim} mirrors the logic of an acoustic Self-Similarity Matrix (SSM), providing a robust signal for recurring sections like the Chorus that might otherwise be diluted by the varying semantic content of individual lines.

The *Base* configuration, which utilizes solely isolated line-level textual input, exhibits the most drastic performance collapse, with a 42.57% drop. This confirms that line-level information in isolation is insufficient for structural analysis. The synergy of all proposed components—particularly the combination of hierarchical semantic context and explicit music-aware cues—is essential for transforming a general-purpose language model into a specialized music structure analyzer capable of achieving robust performance in data-constrained environments.

Table 5.7: Ablation analysis on model components.

Configuration	**Weighted F1**	**Drop (%)**
Proposed Bi-LSTM	**0.7848**	-
w/o Repetition & Position	0.7805	0.55%
w/o Repetition, Position & Section	0.6783	13.57%
w/o Repetition, Position, Section & Window	0.4507	42.57%

5.5 Discussion and Limitations

Although the proposed framework achieves favorable performance and efficiency, several limitations and design considerations remain.

Inference Efficiency and Edge Deployment Feasibility

A critical consideration for real-world deployment is the computational cost during the inference phase. While the core Bi-LSTM classifier is extremely lightweight (5.39 M parameters), the complete inference pipeline for an unseen song requires a forward pass through the RoBERTa encoder (124.65 M parameters) and the calculation of the $O(N^2)$ similarity matrix.

However, we argue that this remains significantly more efficient than audio-based state-of-the-art models for two reasons. First, the sequence length N of lyrics is typically less than 100 lines, whereas audio-based models must process sequences of 10^4 to 10^5 tokens. The quadratic complexity $O(N^2)$ for $N \approx 80$ involves fewer than $6,400$ operations, which is negligible for modern edge AI chips. Second, while RoBERTa is large, the modular nature of our framework allows for the replacement of the base encoder with distilled versions (e.g., DistilRoBERTa or TinyBERT) in extreme resource-constrained scenarios, potentially reducing the parameter count by over 40% while maintaining structural sensitivity.

Pre-processing Requirement

Unlike end-to-end models, the proposed approach requires a pre-processing stage to segment lyrics into line-level units with timestamps. Although this introduces additional overhead, the subsequent inference remains computationally lightweight compared to models operating on high-dimensional acoustic spectrograms. This trade-off is particularly advantageous for Multimedia IoT devices where processing raw audio is energy-prohibitive.

Dependency on Textual Content and Modality Boundaries

Since the model leverages semantic and repetition patterns within lyrics, its effectiveness may vary in tracks with sparse textual content or extended instrumental passages. For music styles with minimal lyrical presence, the model's reliance on text may lead to boundary ambiguity. This is reflected in our results where audio-based models maintained an advantage in detecting *Intros*, which are often characterized by instrumental content rather than lyrical cues. Incorporating lightweight acoustic descriptors may improve structural robustness in these specific transitional segments.

Cognitive Link Between Repetition and Structure

A key assumption of the proposed approach is that lyrical repetition reflects underlying musical structure. This is supported by cognitive studies of music perception, where repetition facilitates segmentation and memory retention. In popular music, sections like the *Chorus* are often characterized by repeated phrases. The proposed repetition features explicitly encode this by measuring semantic similarity across lines. However, in genres with abstract lyrical structures or minimal repetition, this alignment may weaken, potentially limiting the model's performance.

Dataset Scale and Generalization: Quality over Quantity

A primary constraint of this study is the sample size of 148 English popular songs, a direct consequence of the rigorous requirement for triple-synchronized ground truth (audio, line-level timestamps, and structural labels). We strategically prioritized annotation fidelity over raw volume to ensure a precise, gold-standard benchmark. Although the song count is modest, this corpus yields $N = 9662$ individual lyric lines, each serving as a high-density training sample enriched with music-aware positional and repetition features.

To address potential concerns regarding model generalization and overfitting in this data-constrained environment, we provide the following justifications:

- **Cross-Validation Robustness:** We utilized a 5-fold cross-validation strategy, splitting data by song to prevent artist or track leakage. The consistently low standard deviation in weighted F1-scores ($\sigma = 0.0178$) and line-level accuracy indicates that the model is capturing universal structural regularities rather than over-tuning to specific lyrical patterns.
- **Transfer Learning Efficiency:** By leveraging a frozen RoBERTa-base encoder pre-trained on billions of tokens, our framework utilizes massive external linguistic knowledge. The lightweight Bi-LSTM is thus only required to learn the structural mapping logic, which significantly lowers the data volume threshold required for convergence compared to training large-scale audio transformers from

scratch.

- **High-Fidelity Supervision:** Unlike large-scale but weakly-labeled datasets, our synchronized multi-modal corpus prevents ”boundary smear.” The precise temporal grounding ensures that the model learns the exact transitions of the *piecewise functional* musical form, leading to more reliable segmentation.

Our results demonstrate that the semantic-rich and low-dimensional nature of lyrics allows a lightweight model to achieve faster and more stable convergence than high-capacity audio models like SongFormer in scenarios where high-fidelity data is limited. Expanding the dataset to include diverse languages and larger synchronized corpora remains a key direction for future work to further validate the scalability of the proposed lyrics-based MSA framework.

Chapter 6 Conclusion

This study presented a lightweight lyrics-based framework for music structure analysis and examined whether textual signals alone can provide sufficient structural cues for section-level prediction. By constructing a synchronized corpus from the Harmonix Set and DALI datasets, we established a line-level benchmark that aligns lyrics with temporal annotations and structural labels. This setting enabled a direct evaluation of how effectively semantic and positional information can support music structure modeling without relying on computationally expensive acoustic representations.

Our results suggest that explicitly incorporating music-aware priors into a compact Bi-LSTM architecture is an effective strategy for this task. The combination of hierarchical semantic embeddings, repetition-aware features, and positional cues consistently improved structural discrimination, particularly for dominant functional sections such as *Chorus* and *Verse*. Compared with both LongFormer and SongFormer baselines, the proposed framework achieved stronger overall performance while requiring substantially fewer computational resources. These findings indicate that, at least for modern English popular music, lyrics contain richer structural information than is often assumed.

From a practical perspective, the reduction in memory consumption (20.5× lower peak GPU usage than SongFormer) highlights the feasibility of deploying lyrics-driven structure analysis in resource-constrained environments. This advantage may be particularly relevant for edge-oriented MIR applications where latency and hardware limitations restrict the use of large-scale audio transformers.

At the same time, the current framework has several limitations. Its effectiveness depends on the availability of sufficiently informative lyrical content, and performance remains weaker for instrumental-heavy transitions such as *Outro*. In addition, it remains unclear how well the observed structural patterns generalize to genres with less repetitive lyrical organization or to multilingual settings.

Future work will integrate lightweight acoustic features with the proposed lyrics-based framework to improve instrumental section recognition while preserving computational efficiency.

Code and Data Availability

To facilitate reproducible research, the implementation of the proposed framework, including model training and evaluation codes, trained model checkpoints, and experimental configurations, has been made publicly available through GitHub:

`https://github.com/RennyYo/Multi-Scale-Semantic-Tagging-for-Lyrics-Based-Music-Structure-Analysis-Using-Lightweight-Bi-LSTM`

The synchronized dataset constructed from the intersection of the DALI v2.0 dataset and the Harmonix Set is available upon reasonable request, subject to the licensing terms of the original datasets.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used AI-assisted language editing tools to improve the clarity and presentation of this manuscript. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication. The scientific contributions, experimental design, data collection, analysis, and conclusions are entirely the authors' own work.

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